steady_drift_calm

equity strategy | family: equity-longonly | status: research | derived from `steady_drift`

What this strategy does

Steady Drift with the risk throttle at 8% volatility (smaller drawdowns, a little less return)

Why it should work. The paid leg of PEAD is the long side of the top surprise decile; size by calmness, never borrow, and keep book risk roughly constant so the edge is not swamped by market drawdowns.

Design notes (from the strategy's source file).

Steady Drift -- a long-only, never-borrowing PEAD book with a constant-risk throttle (research).

What it does * Buys only the best earnings surprises (top decile of the point-in-time SUE rank) in stocks that trade at least $2M a day, holds each for exactly 60 trading sessions, and never sells short. * Sizes each position by how calm the stock is (inverse idiosyncratic variance, relative to the typical stock of the past year), at most 2% of NAV per name and 5% of the stock's daily dollar volume. Positions are admitted in announcement order, so no decision depends on events that have not happened yet. * Total stock cost basis is capped at 70% of NAV; the rest is cash earning the Treasury-bill rate (not the index). * A risk throttle scales the whole book down when its recent volatility exceeds the target (never above 100% invested, so it can only reduce risk, never borrow).

Why: after the methodology fixes, every market-neutral book collapses to 1-3% a year once short collateral is counted (they tie up 1.2-5.6x NAV), and the options edge disappears at real bid/ask. What survives is the long side of the surprise signal; docs/STEADY_DRIFT.md and docs/META_STRATEGY.md carry the evidence and the shuffled-surprise placebo.

STATUS research: the design was chosen after seeing 1996-2013; a shuffled-surprise test shows about half of the excess return over the cap-weighted index is the event universe (small/mid-cap, low-vol), not surprise skill; the T-bill series is typed from memory (pead/core/rates.py).

parameter	value
base_unit_fraction	0.004
max_gross_leverage	0.7
liquidity_cap_frac	0.05
return_basis	raw
cap_mode	chronological
exposure_cap	open_book
beta_overlay	0.0
hold_sessions	60
long_only_cash	True
risk_target_vol	0.08
name_cap_frac	0.02

Performance

metric	steady drift calm	market (CRSP VW)

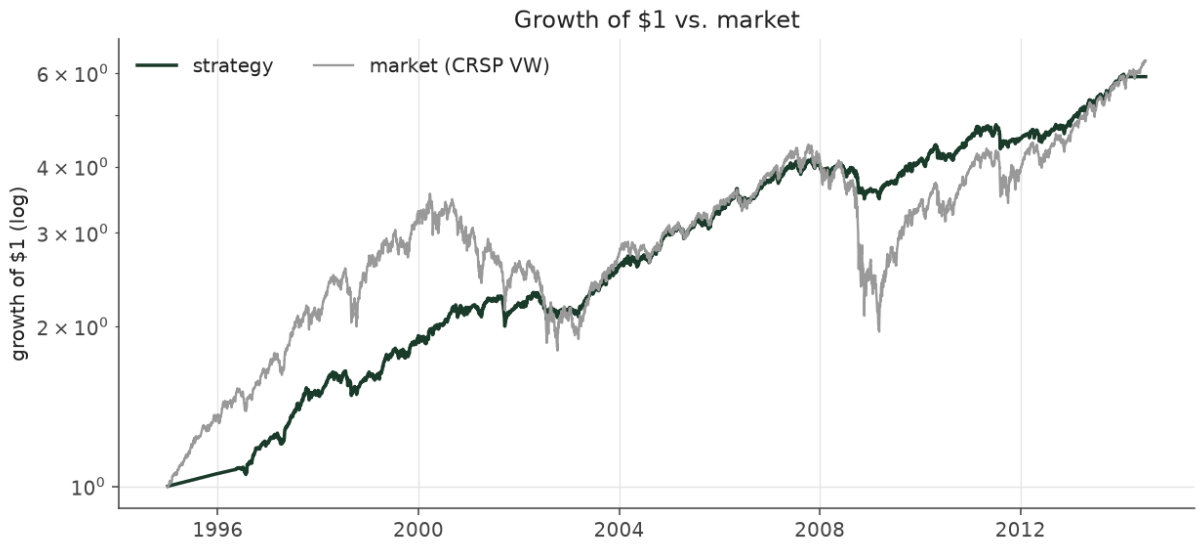
Annualized return	9.55%	9.94%
Annualized volatility	7.81%	19.35%
Sharpe (NAV-pct)	1.21	0.59
Sortino	1.74	
Max drawdown	-15.9%	-55.5%
Calmar	0.60	
Final NAV ($10M start)	$59,076,029	
Worst day	-4.33%	
% positive days	59.5%	
Sharpe 1996-2001	1.61	
Sharpe 2002-2007	1.30	
Sharpe 2008-2013	0.79	

Correlation of daily returns with the market: **0.84**.

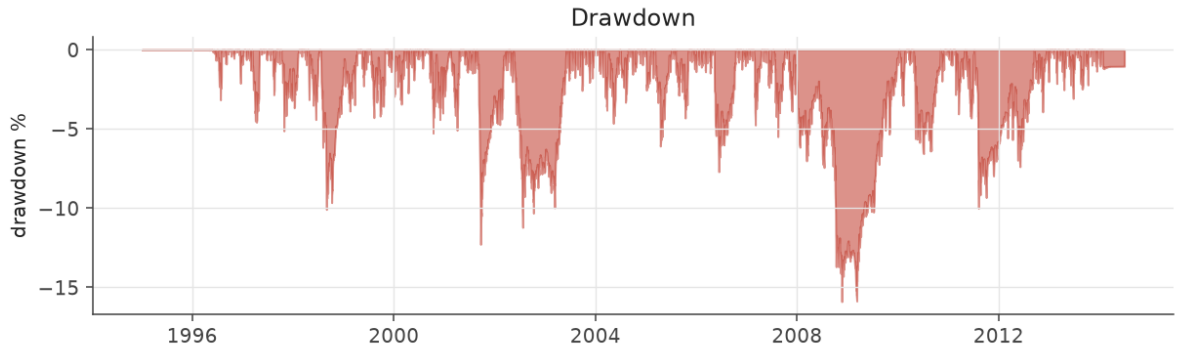
• best Sharpe among equity: `calm_rank_tilt` (1.47); this one: 1.21

• best return among equity: `market_boost_max` (0.116); this one: 0.0955

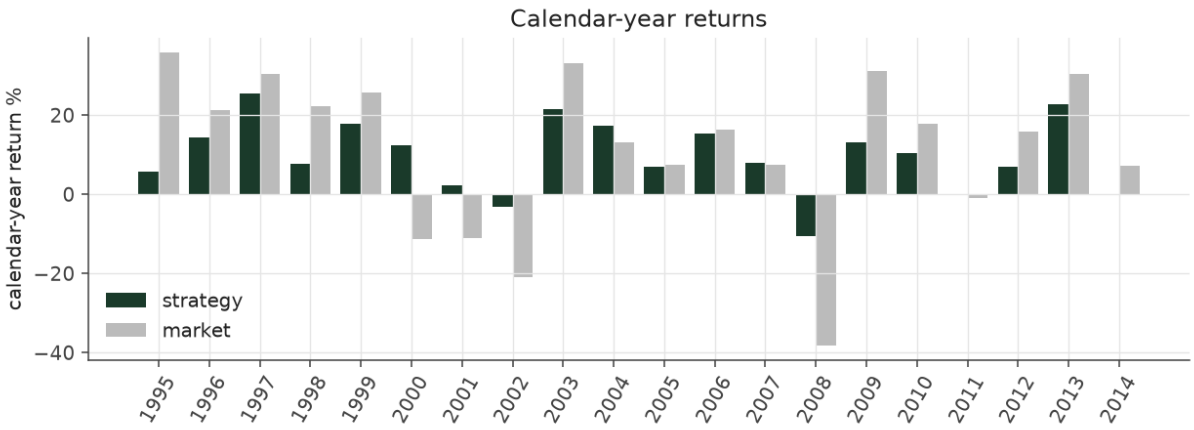
• best Calmar among equity: this strategy; this one: 0.601



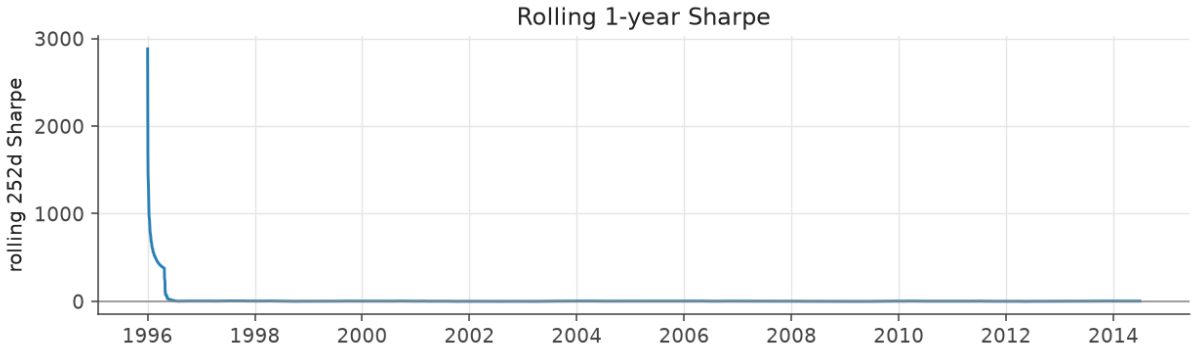
NAV (log scale)



Drawdown from running peak

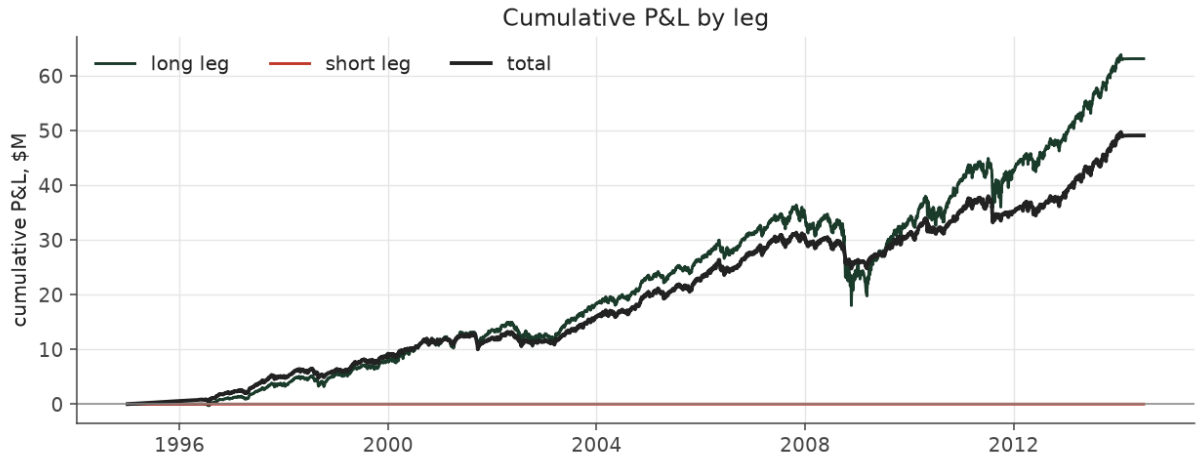


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

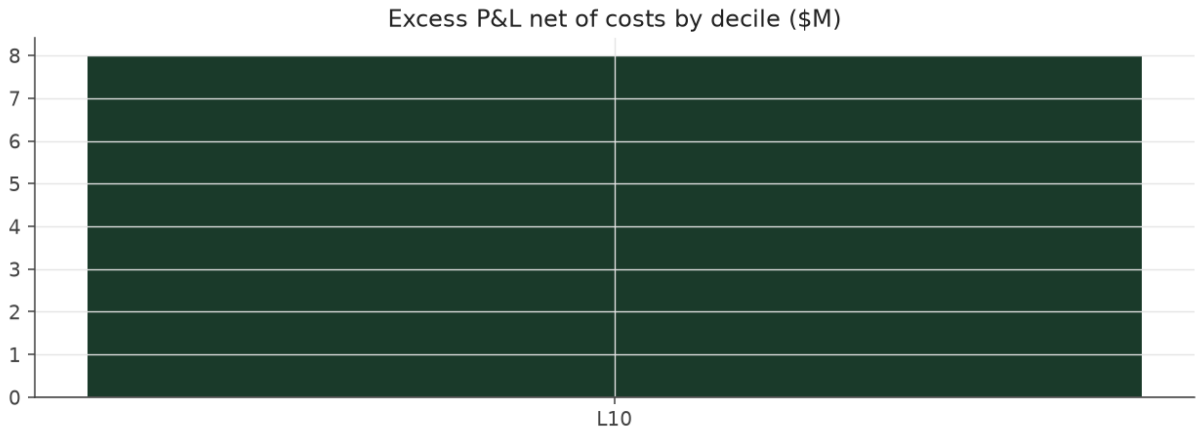
	long leg	short leg
P&L ($M, before costs)	63.10	0.00
P&L after costs ($M)	60.97	0.00
of which common (average-event) component ($M)	21.72	0.00
of which excess / PEAD component ($M)	10.14	0.00
excess minus costs ($M)	8.00	0.00
gross return per $ traded (bp)	405.42	-
positions in the money (hit rate)	0.64	-
stand-alone Sharpe (leg only)	0.86	-

Daily P&L correlation between the two legs: **nan**.

• The two legs' daily P&L are nan correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 0.86 long / nan short vs 1.21 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	10	8,467	1,556.35	60.97	10.14	2.13	8.01	100.0%	51.43



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	65	3.02	0.08	-0.08	0.01	-0.08	-1.0%	-277.25
2	803	55.91	3.76	1.51	0.14	1.37	17.1%	245.22
3	1,772	193.88	9.65	2.94	0.37	2.56	32.0%	132.08
4	2,711	462.97	17.98	3.94	0.70	3.25	40.5%	70.09
5	3,116	840.56	29.50	1.82	0.91	0.91	11.4%	10.84

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
2	471	53.85	2.15	0.37	0.16	0.21	2.6%	38.51
3	1,984	264.19	13.93	5.48	0.53	4.96	61.9%	187.55
4	2,785	507.59	19.31	4.01	0.71	3.30	41.3%	65.08
5	3,227	730.71	25.57	0.27	0.73	-0.46	-5.8%	-6.31

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
41-60d	8,467	1,556.35	60.97	10.14	2.13	8.01	100.0%	51.43

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	2,988	231.12	13.05	6.17	0.32	5.86	73.1%	253.35
2002-07	2,537	527.79	17.84	5.83	0.70	5.13	64.1%	97.25
2008-13	2,942	797.44	30.07	-1.87	1.12	-2.98	-37.3%	-37.41

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	2,075	263.14	11.67	2.37	0.36	2.01	25.1%	76.40
Chems	200	57.34	2.72	1.56	0.08	1.48	18.5%	257.75
Durbl	200	38.65	0.70	-0.57	0.05	-0.62	-7.8%	-160.56
Enrgy	241	39.81	2.21	0.92	0.05	0.86	10.8%	217.00
Hlth	664	127.99	3.97	-0.36	0.16	-0.52	-6.5%	-40.87
Manuf	1,029	223.82	7.01	2.33	0.30	2.03	25.3%	90.64
Money	1,258	334.50	12.98	0.66	0.47	0.19	2.4%	5.68
NoDur	405	85.35	3.72	0.76	0.11	0.65	8.1%	75.77
Other	1,068	162.42	7.19	2.92	0.24	2.67	33.4%	164.68

Shops	862	123.90	4.58	0.03	0.17	-0.15	-1.9%	-12.01
Telcm	187	19.62	0.96	0.14	0.03	0.12	1.4%	58.95
Utils	278	79.82	3.25	-0.60	0.11	-0.71	-8.9%	-89.03

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	3,155	224.85	7.38	0.68	0.29	0.38	4.7%	16.89
low vol	2,478	841.05	35.34	6.40	1.16	5.24	65.5%	62.31
mid vol	2,704	482.63	18.02	2.96	0.66	2.30	28.8%	47.70

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	9.55%	0.00%
0.5%	0.0	9.55%	0.00%
1.0%	0.0	9.55%	0.00%
2.0%	0.0	9.55%	0.00%
5.0%	0.0	9.55%	0.00%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.